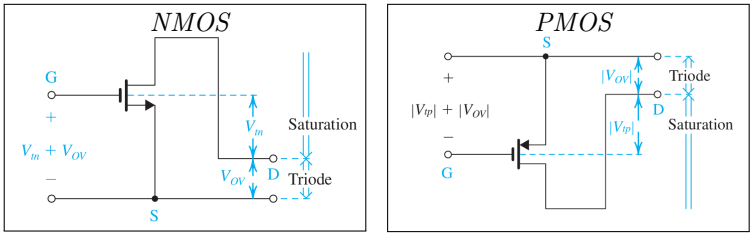
# ECE 303 Formula Sheet

Chukwugozie Okeke

## MOSFET Transistor

*Operating Modes*

## MOSFET Small Signal Model



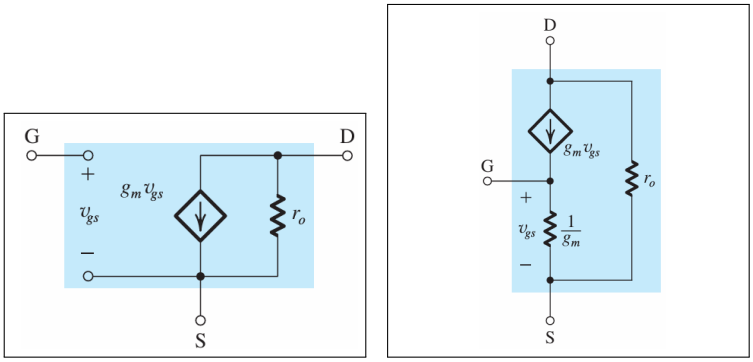

1. Cutoff: $v_{GS} \leq V_{OV}$
2. Triode: $v_{GS} \geq V_{tn}$ AND $v_{DS} \leq V_{OV} \implies v_{GD} \geq V_{tn}$
3. Saturation: $v_{GS} \geq V_{tn}$ AND $v_{DS} \geq V_{OV} \implies v_{GD} \leq V_{tn}$

(Satur.): $I_D = I_S = \frac{1}{2}k_n\left[(V_{GS} - V_{tn})^2(1 + \lambda V_{DS})\right], \qquad I_G = 0$

(Triode): $I_D = \frac{1}{2}k_n\left[2(V_{GS} - V_{tn})V_{DS} - V_{DS}^2\right]$

$k_n = k_n^{'}\frac{W}{L} = \mu_n C_{ox}\frac{W}{L} \qquad \lambda = \frac{1}{V_A}$

## MOSFET Small Signal Model




$g_m = k_n|V_{OV}| = \frac{2I_D}{|V_{OV}|} \qquad r_o = \frac{1}{\lambda I_D} = \frac{V_A}{I_D} \qquad v_{id} << 2V_{OV}$

## MOSFET Differential Amplifiers

$R_{id} = \infty \qquad A_d = -g_m(R_D \parallel r_o)$

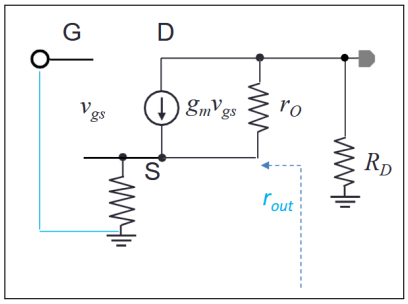
*MOSFET Differential Pair With Source Degeneration*

$A_v = -g_{m-eff}R_D \qquad g_{m-eff} = \frac{g_m}{1 + g_m R_s}R_d$

$r_{out} = R_S + r_o(1 + g_m R_S)$

*MOSFET Common Gate Input Resistance*

$r_{in} = \frac{r_o + R_D}{1 + g_m r_o}$




*Input Common-mode Range*

$V_{CMmax} = V_{tn} + V_D = V_{DD} + V_{tn} - \frac{I}{2}R_D$

$V_{CMmin} \approx V_{GS} + V_{CSmin} - V_{SS}$

*MOSFET Common-Mode*

$CMRR = \frac{|A_d|}{|A_{cm}|} \quad A_{cm} = -\frac{R_D}{2R_{SS}}\left(\frac{\Delta R_D}{R_D} + \frac{\Delta g_m}{g_m}\right) \quad A_d = g_m R_D$

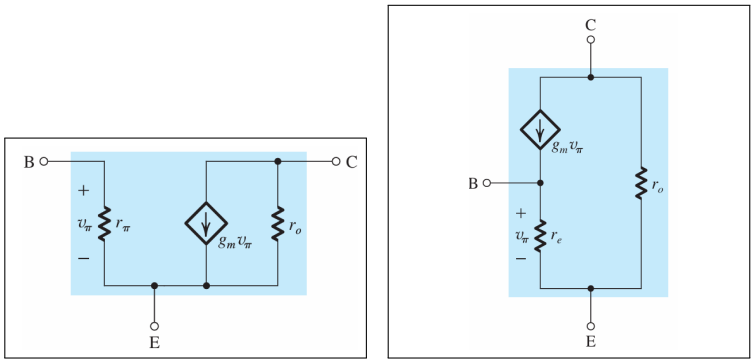
## BJT Transistor

*Operating Modes*

1. Cutoff: EBJ Reverse, CBJ Reverse
2. Active: EBJ Forward, CBJ Reverse
   $\implies v_{BE} \geq 0.4V, v_{CB} \geq -0.3V$
3. Saturation: EBJ Forward, CBJ Forward

$i_c = I_s e^{\frac{v_{BE}}{V_T}}\left(1 + \frac{v_{CE}}{V_A}\right) \quad i_c = \alpha i_e = \beta i_b \quad i_e = i_c + i_b = (\beta + 1)i_b$

## BJT Small Signal Model




$g_m = \frac{I_c}{V_T} \qquad r_o = \frac{V_A}{I_C^{'}} \qquad r_\pi = \frac{V_T}{I_B} = \frac{\beta}{g_m} \qquad r_e = \frac{V_T}{I_E} = \frac{\alpha}{g_m}$

$r_\pi = (\beta + 1)r_e \qquad \alpha = \frac{\beta}{\beta + 1} \qquad \beta = \frac{\alpha}{1 - \alpha} \qquad v_{id} << 2V_T$

## BJT Differential Amplifiers

$A_d = g_m(R_C \parallel r_o) \qquad R_{id} = 2r_\pi$

*BJT Differential Pair With Emitter Resistances*

$$i_e = \frac{v_{id}}{2(r_e + R_e)} \qquad R_{id} = 2(\beta+1)(r_e + R_e) \qquad A_d = \frac{\alpha R_c}{r_e + R_e}$$

$$r_{out} = (R_E \parallel r_\pi) + r_o(1 + g_m(R_E \parallel r_\pi))$$

*Input Common-mode Range*

$$V_{CMmax} \approx V_C + 0.4 = V_{CC} - \alpha\frac{I}{2}R_C + 0.4$$

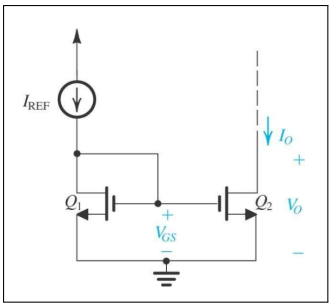
$$V_{CMmin} \approx -V_{EE} + V_{CSmin} + V_{BE}$$

*BJT Common-Mode*

$$CMRR = \frac{|A_d|}{|A_{cm}|} \quad A_{cm} = -\frac{R_C}{2R_{EE}}\left(\frac{\Delta R_C}{R_C} + \frac{\Delta\beta}{\beta}\right) \quad A_d = g_m R_C$$

. . . . . .

**Current Mirrors**




Resistive Reference $\implies I_o = I_{ref} = \dfrac{V_{DD} - V_{GS}}{R_{ref}}$

. . . . . .

**Frequency Response Basics**

*OCTC (Low-Pass Pole)*

1. Draw the s.s. circuit
2. S.c all LF (or series) caps and indep. AC voltage sources, o.c. all indep. AC current sources
3. Select HF cap, replace all others with o.c's.
4. Find resistance seen by this cap. then find $\tau_n = R_n C_n$, repeat for other caps
5. Corner frequency is $\omega_p = 1/(\tau_1 + \tau_2 + \tau_3 + ...)$

$$\frac{v_o}{v_i} = \frac{A}{1 + s/\omega_p}$$

*SCTC (High-Pass Pole)*

1. Draw the s.s. circuit
2. O.c all HF (or series) caps and indep. AC current sources, s.c. all indep. AC voltage sources
3. Select LF cap, replace all others with s.c's.
4. Find resistance seen by this cap. then find $\tau_n = R_n C_n$, repeat for other caps
5. Corner frequency is $\omega_p = 1/\tau_1 + 1/\tau_2 + 1/\tau_3 + ...$

$$\frac{v_o}{v_i} = \frac{As}{s + \omega_p}$$

*Dominant Pole*

Low Freq. Caps (SCTC): $\implies f_{p2} \geq 4f_{p3}$.

If not, use $f_L = \sqrt{f_{P1}^2 + f_{P2}^2 + f_{P3}^2 - 2f_Z^2}$

High Freq. Caps (OCTC): $\implies f_{p3} \geq 4f_{p2}$.

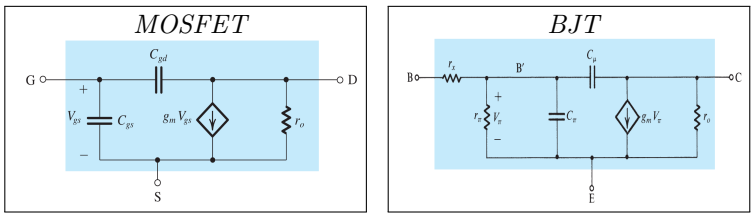
If not, use $f_H = \sqrt{1/f_{P1}^2 + 1/f_{P2}^2 + 1/f_{P3}^2 - 2/f_Z^2}$

. . . . . .

**Frequency Response of Amplifiers**

1. Draw s.s. circuit
2. Identify LF and HF Caps
3. Find Midband gain (i.e. s.c all HP caps., o.c. all LP caps)
4. HP Pole (SCTC) and LP Pole (OCTC)

*MOSFET/BJT HF Models*




*Miller's Theorem*

$$Z_1 = \frac{Z_F}{1 - A_v} \qquad Z_2 = \frac{Z_F}{1 - 1/A_v} \qquad A_v = \frac{v_2}{v_1}$$

$$C_1 = (1 - A_v)C \qquad C_2 = (1 - 1/A_v)C \qquad A_v = \frac{v_2}{v_1} = -g_m R_{Leq}$$

For BJT's: $C_{in} = C_\pi + (1 + |A_V|)C_\mu \quad C_{out} = (1 + 1/|A_v|)C_\mu$

For MOS's: $C_{in} = C_{gs} + (1 + |A_V|)C_{gd} \quad C_{out} = (1 + 1/|A_v|)C_{gd}$

*Common Source/Common-Emitter Amplifiers*

$$s_z = \frac{g_m}{C_{gd}} \implies f_z = \frac{g_m}{2\pi C_{gd}}$$

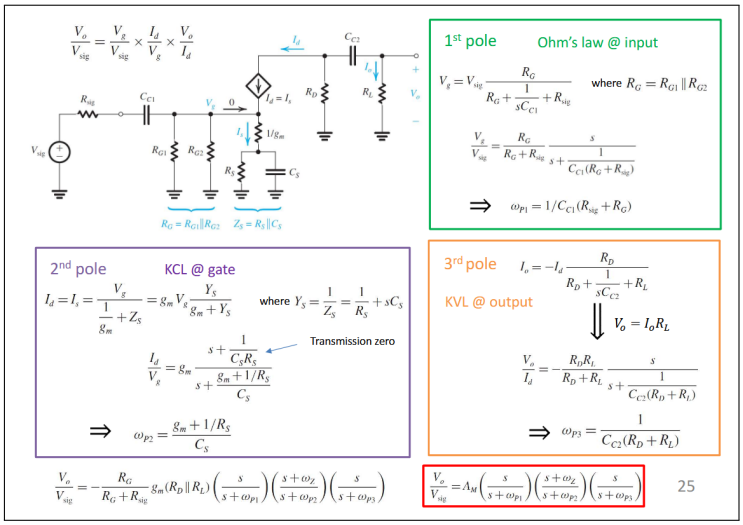
(MOS) $f_T = \dfrac{g_m}{2\pi(C_{gs} + C_{gd})}$ (BJT) $f_T = \dfrac{g_m}{2\pi(C_\pi + C_\mu)}$

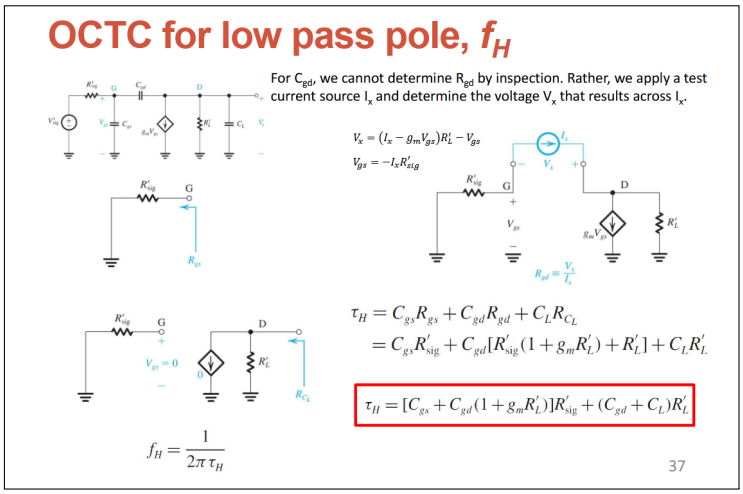
$$\frac{V_{out}}{V_{in}} = \frac{-g_m R_{Leq}(1 - \frac{sC_{gd}}{g_m})}{(1 + sC_{gd}R_{Leq})(1 + sC_{in}R_{th})}$$

*Common Gate Amplifier*

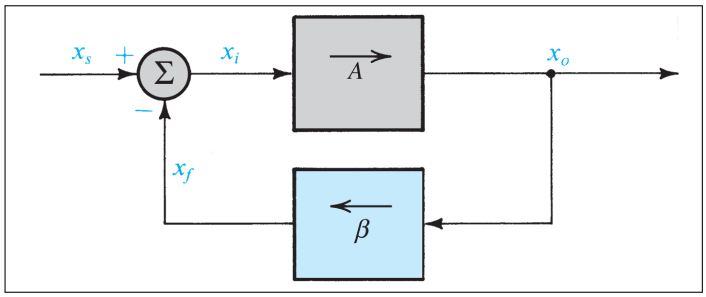
$$A_M = \frac{g_m R_{Leq}}{1 + g_m R_{sig}}$$

. . . . . .




**Feedback**




$$A_f = \frac{x_o}{x_s} = \frac{A}{1+A\beta} \approx \frac{1}{\beta} \qquad \beta = \frac{x_f}{x_o} \qquad A\beta = -\frac{x_r}{x_o}$$

Amt. of Feedback $\implies (1+A\beta)$ Loop Gain $\implies A\beta$

*Output Impedance of MOSFET from Drain (Common-Gate)*

$$R_{out} = r_o + (1+g_m r_o)R_s \approx (g_m r_o)R_s$$

*Connection Types*

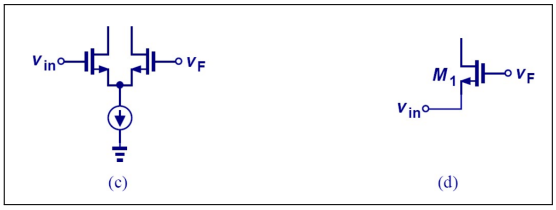
Series (mix v, measure i) $\implies R_{in/out} = R_{if/of} - R_{sig/L}$

Shunt (measure v, mix i) $\implies R_{in/out} = \dfrac{1}{1/R_{if/of} - 1/R_{sig/L}}$

*Open-loop Networks/Parameters*

1. For A, identify feedback type, duplicate feedback: shunt $\rightarrow$ short, series $\rightarrow$ sever. Calculate gain, $R_o$ and $R_i$. Use table to calculate $R_{if}$ and $R_{of}$. Be careful to account for MOSFET $i_G = 0$ when calculating parallel resistors (also, $R_{in} = \infty$)
2. For $\beta$ draw feedback network, calculate $x_f/x_i$ depending on network type (ex. for shunt-shunt (i-i) $x_f = i_f, x_o = v_o$). Note: When drawing beta network, currents should point INTO network ($A\beta \geq 0$ for negative feedback)

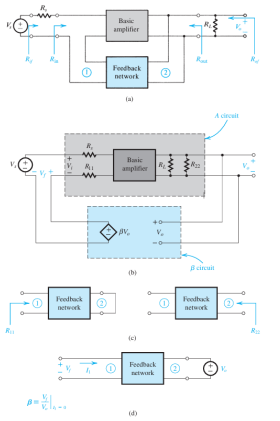
*Series Connections Examples*



## Feedback Topologies Summary Table

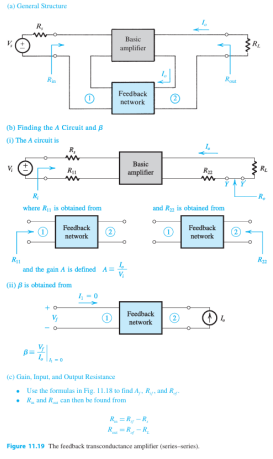
**Table 11.2** Summary of Relationships for the Four Feedback-Amplifier Topologies

| Feedback Amplifier | Feedback Topology | $x_i$ | $x_o$ | $x_f$ | $x_s$ | $A$ | $\beta$ | $A_f$ | Source Form | Loading of Feedback Network Is Obtained: At Input | Loading of Feedback Network Is Obtained: At Output | To Find $\beta$, Apply to Port 2 of Feedback Network | $R_{if}$ | $R_{of}$ | Refer to Figs. |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Voltage | Series–shunt | $V_i$ | $V_o$ | $V_f$ | $V_s$ | $\frac{V_o}{V_i}$ | $\frac{V_f}{V_o}$ | $\frac{V_o}{V_s}$ | Thévenin | By short-circuiting port 2 of feedback network | By open-circuiting port 1 of feedback network | a voltage, and find the open-circuit voltage at port 1 | $R_i(1+A\beta)$ | $\frac{R_o}{1+A\beta}$ | 11.12 11.14 |
| Current | Shunt–series | $I_i$ | $I_o$ | $I_f$ | $I_s$ | $\frac{I_o}{I_i}$ | $\frac{I_f}{I_o}$ | $\frac{I_o}{I_s}$ | Norton | By open-circuiting port 2 of feedback network | By short-circuiting port 1 of feedback network | a current, and find the short-circuit current at port 1 | $\frac{R_i}{1+A\beta}$ | $R_o(1+A\beta)$ | 11.25 11.26 |
| Transconductance | Series–series | $V_i$ | $I_o$ | $V_f$ | $V_s$ | $\frac{I_o}{V_i}$ | $\frac{V_f}{I_o}$ | $\frac{I_o}{V_s}$ | Thévenin | By open-circuiting port 2 of feedback network | By open-circuiting port 1 of feedback network | a current, and find the open-circuit voltage at port 1 | $R_i(1+A\beta)$ | $R_o(1+A\beta)$ | 11.18 11.19 |
| Transresistance | Shunt–shunt | $I_i$ | $V_o$ | $I_f$ | $I_s$ | $\frac{V_o}{I_i}$ | $\frac{I_f}{V_o}$ | $\frac{V_o}{I_s}$ | Norton | By short-circuiting port 2 of feedback network | By short-circuiting port 1 of feedback network | a voltage, and find the short-circuit current at port 1 | $\frac{R_i}{1+A\beta}$ | $\frac{R_o}{1+A\beta}$ | 11.22 11.23 |



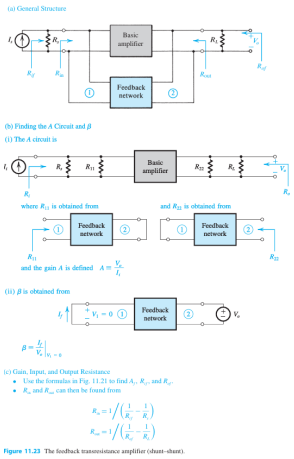

*Series-Shunt (v-v)*

Figure 11.14 (a) Block diagram of a practical series–shunt feedback amplifier. (b) The circuit in (a) represented by the ideal structure of Fig. 11.12(a). (c) Definition of $R_{i1}$ and $R_{i2}$. (d) Determination of the feedback factor $\beta$. (e) The $A$ circuit, showing the open-loop resistances $R_i$ and $R_o'$.



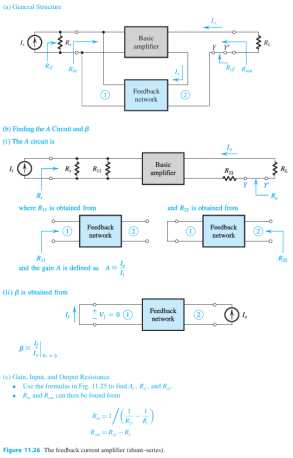

*Series-Series (i-v)*

Figure 11.19 The feedback transconductance amplifier (series–series).




*Shunt-Shunt (v-i)*

Figure 11.23 The feedback transresistance amplifier (shunt–shunt).




*Shunt-Series (i-i)*

Figure 11.26 The feedback current amplifier (shunt–series).